ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

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Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer:

1. There are 594 documents with the category sci.med.
2. Because it enables us to focus on data that we're currently investigating and reveal topic-specific word combinations.

AI used for this answer: Yes

If yes, how:

Guidance and concept explanation.

External references (links):

https://pandas.pydata.org/docs/user_guide/indexing.html#boolean-indexing

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Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer:

Dropping 40% of the documents with missing categories would be a bad idea, because we would lose their potentially useful text that we could've used. One alternative is to ignore the missing value during analysis.

AI used for this answer: Yes

If yes, how:

Guidance and concept explanation.

External references (links):

https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.dropna.html

https://scikit-learn.org/stable/auto_examples/text/plot_document_clustering.html

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Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer:

1. The proportions are similar but are not a perfect match.

Full dataset

Category	soc.religion.christian	sci.med	comp.graphics	alt.atheism	Total
Number	599	594	584	480	2257
Proportion	26.5%	26.3%	25.9%	21.3%	100%

Sample (N=1000)

Category	soc.religion.christian	sci.med	comp.graphics	alt.atheism	Total
Number	248	260	278	214	1000
Proportion	24.8%	26.0%	27.8%	21.4%	100%

2. Sampling an imbalance dataset could result in a rare category getting few sampled documents, or even none.

AI used for this answer: Yes

If yes, how:

Concept explanation and answer checking.

External references (links):

https://scikit-learn.org/stable/modules/cross_validation.html#cross-validation-iterators-with-stratification-based-on-class-labels

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Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?
2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer:

1. The number of features (columns) increased from 35482 to 35788.
2. Adding back the stop words would add more columns and non-zero elements to the matrix, causing it to be less sparse.

AI used for this answer: No

If yes, how:

External references (links):

Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

1. As the threshold increases, the size of the vocabulary decreases.

Threshold	Number of terms kept (keep == True)
0.0	11676
0.01	3714
0.05	1288

2. Words/terms that are extremely rare like ”020701tan102” have close-to-zero variances and are eliminated first.
3. Because features with very little variation don't provide much information to distinguish documents and categories apart. Eliminating those features would hardly reduce the model’s accuracy.

AI used for this answer: Yes

If yes, how: Answer checking for subquestions 1 and 2, concept explanation for subquestion 3.

External references (links):

https://scikit-learn.org/stable/modules/feature_selection.html

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Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
2. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?
3. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer:

1. The top-correlated words are a mix of graphics terminology (e.g. vga, video, card) and generic words (e.g. thanks, hi).
2. Pearson correlation measures linear association using the word counts. Spearman correlation measures monotonic association based on the ranks of the word counts.

3. It could happen when a word has higher counts in graphics documents than in other documents, but a few documents repeat it very often.

AI used for this answer: Yes

If yes, how: Guidance and concept explanation

External references (links):

https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.pearsonr.html

https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.spearmanr.html

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Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup:

minSup	Runtime (seconds)	Pattern Count
9	0.117	710
3	1.619	9547

Answer (2) — Top-K k vs MaxFPGrowth minSup:

Top-K FAE finds frequent patterns with the highest support counts, where `k` controls how many patterns to find. MaxFPGrowth finds maximal frequent patterns whose support count is equal or greater than a specified number `minSup`.

Answer (3) — Maximal vs standard frequent patterns:

Maximal frequent pattern mining returns frequent patterns that have no larger frequent pattern supersets, and omits smaller standard frequent subsets, making maximal frequent patterns more concise.

AI used for this answer: Yes

If yes, how: Concept explanation and answer checking.

External references (links):

https://data-mining.philippe-fournier-viger.com/what-is-a-maximal-itemset/

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer:

1. Because each document in X_counts has thousands of dimensions (words). The most we can do is 3 dimensions with a 3D scatter plot.
2. We apply **dimensionality reduction algorithms**, such as PCA, t-SNE, and UMAP.

AI used for this answer: Yes

If yes, how: Concept explanation on "dimension"

External references (links):

https://www.geeksforgeeks.org/machine-learning/dimensionality-reduction/

Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?

3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer:

1. PCA shows a very heavy overlap at the center. t-SNE spreads the documents further, but documents are still mixed around the center. UMAP makes a clearer grouping between the 4 document categories, but even so, there is still some overlap around the center.
2. PCA uses linear projection on directions that preserves the most variance, but those directions may not separate the document categories. Meanwhile, t-SNE and UMAP keep similar documents near one another.
3. Because in t-SNE, a large physical distance doesn't necessarily reflect a large difference between documents, because t-SNE emphasizes local similarities rather than preserving distances.

AI used for this answer: Yes

If yes, how: Concept explanation and guidance.

External references (links):

https://scikit-learn.org/stable/modules/decomposition.html#pca

https://umap-learn.readthedocs.io/en/latest/how_umap_works.html

https://distill.pub/2016/misread-tsne

Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

Answer:

1. There are 4 binary columns to represent 4 categories.
2. Algorithms like linear regression or KNN can falsely assume that the numerical distances between category numbers represent similarity. For example, they may see category 3 as twice as far away from category 1 than category 2.
3. One-hot binarization assigns a separate binary indicator per category, thus removing the numerical scale between categories. The Euclidean distance between any two categories would always be $\sqrt{2}$.

AI used for this answer: Yes

If yes, how: Concept explanation

External references (links):

https://www.geeksforgeeks.org/machine-learning/ml-one-hot-encoding/
